

CT Jobportal - Comprehensive Testing Document

1. Overview

This document outlines the testing strategy, test cases, and responsibilities for the CT Jobportal platform. The testing process is divided into Technical and Non-Technical testing phases to ensure the platform is secure, performant, functionally correct, and visually premium.

2. Team Assignments

- **Technical Testing Team:** Karthika, Pallavan
 - **Non-Technical (Functional/UI) Testing Team:** Babila
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3. Technical Testing Plan (Assigned to: Karthika & Pallavan)

Focus Areas: Backend APIs, Database integrity, Integrations (Razorpay, Webhooks), Security, and Performance.

3.1. API & Backend Testing

- **Authentication & Authorization:**
 - Verify JWT token generation, expiration, and refresh logic.
 - Test Role-Based Access Control (RBAC) across Admin, College, Company, and Jobseeker routes.
 - Ensure protected routes return `401 Unauthorized` or `403 Forbidden` when accessed without appropriate credentials.
- **CRUD Operations:**
 - Test all endpoints for Users, Jobs, Applications, Subscriptions, and Career Counselling requests.
 - Validate request body parsing and error handling for missing or malformed data.
- **Rate Limiting & Throttling:**
 - Test API rate limits on public endpoints (e.g., login, registration, free assessments) to prevent brute-force attacks.

3.2. Integration Testing

- **Razorpay Payments & Subscriptions:**
 - Test the end-to-end payment flow for Company and College subscriptions.
 - Validate the `verifySubscriptionPayment` and `verifyPayment` controllers.
 - Test coupon logic: Ensure percentage-based discounts are accurately calculated and usage counts are incremented.
 - Test Webhook processing for successful payments, failed payments, and subscription auto-renewals.
- **External Communications:**
 - Verify transactional email deliveries (Registration, Password Reset, Interview Invites).
 - Test CSV upload logic for Bulk Student Import (College module) - check for parsing errors, duplicate handling, and data mapping.

3.3. Security & Data Validation

- **Vulnerability Checks:**
 - Test for SQL/NoSQL Injection vulnerabilities in search bars and data input fields.
 - Ensure Cross-Site Scripting (XSS) prevention on User Inputs (e.g., Job descriptions, Blog posts, Resumes).
- **Data Integrity:**
 - Verify Mongoose schema validations (Required fields, unique constraints like email, phone numbers).
 - Test the `isActiveSeat` boolean logic for Company team members to strictly enforce subscription seat limits.

3.4. Performance & Load Testing

- **Database Optimization:**
 - Test query execution times, especially for complex aggregations (e.g., TPO Placement Reports, Admin Dashboards).
 - Ensure proper indexing on frequently searched fields (emails, roll numbers, job titles).
 - **Concurrent Users:**
 - Simulate concurrent job applications and mock interview requests to ensure the server handles load gracefully without race conditions.
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4. Non-Technical Testing Plan (Assigned to: Babila)

Focus Areas: UI/UX aesthetics, Functional workflows, Cross-browser compatibility, and User journeys.

4.1. UI/UX & Aesthetics (Premium Design Verification)

- **Visual Consistency:**
 - Verify the Glassmorphism effects, gradient accents, and typographic hierarchy across all dashboards (Admin, College, Company, Jobseeker).
 - Check the newly implemented "Lens" cursor animation for smooth, physics-based interactions and hover states.
- **Component Rendering:**
 - Ensure metric cards, header containers, and sidebar widgets align perfectly.
 - Test Blog UI components: Dark-themed footers, SVG iconography, color-coded comparison tables, and HTML5 accordion FAQs.

4.2. User Role Workflows (Functional Testing)

- **College / TPO Portal:**
 - **Onboarding:** Test the proof-of-identity upload and admin verification flow.
 - **Student Management:** Test the dynamic, searchable student list (search by name, email, roll number, phone).
 - **Bulk Actions:** Verify the CSV upload feature properly displays success/error logs and enforces plan-based limits. Test bulk status updates for students.
 - **Passkey:** Verify the read-only Principal Executive Passkey dashboard works correctly and displays live placement monitoring.
- **Company / Recruiter Portal:**
 - **Team Management:** Test the toggle mechanism for assigning the single available seat (`isActiveSeat`) to specific team members.
 - **Subscriptions & Coupons:** Verify the UI correctly updates the total payable amount when applying a discount coupon during checkout.
 - **Interviews:** Test the Mock Interview creation and scheduling workflows.
- **Jobseeker Portal:**
 - **Career Counselling:** Test the Edit/View modes for pending counselling sessions. Ensure jobseekers can clearly see the distinction between "Cancelled" and "Rejected" statuses.
 - **Assessments:** Complete the Free Assessment flow and verify results processing.
 - **Job Applications:** Test applying to jobs, uploading resumes, and tracking application statuses.
- **Admin Dashboard:**
 - Verify global metrics, user approval workflows (for colleges/companies), and dispute/refund logic for cancelled counselling sessions.

4.3. Responsive Design & Cross-Browser Testing

- Test all platform modules on Desktop, Tablet, and Mobile devices.
- Ensure UI elements like step arrows in `ExplainerTabs` are perfectly centered and do not break on smaller screens.
- Verify cross-browser functionality on Chrome, Safari, Firefox, and Edge.

4.4. Content & SEO Testing

- Verify SEO tags (Title tags, Meta descriptions) on public pages (Home, Blogs, Free Assessment).
- Ensure proper formatting, standardized list indentation, and functioning hyperlinks within Blog posts.

5. Bug Reporting Protocol

When defects are found, they should be documented in the issue tracker using the following format:

- **Title:** Brief description of the issue.
- **Environment:** (e.g., Production, Staging / Chrome 114 / Windows 11).
- **Module/Role:** (e.g., Company Subscription Checkout).
- **Steps to Reproduce:** Numbered list of actions leading to the bug.
- **Expected Result:** What should have happened.
- **Actual Result:** What actually happened.
- **Attachments:** Screenshots, screen recordings, or API error logs.